

FOSSIL FUEL EFFECTS ON THE GLOBAL CARBON CYCLE

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INTRODUCTION

A radiative energy balance model describes the exchange of energy within the Earth's system. In this assignment we developed a simple radiative energy balance model for the Earth, then explored the effects of volcanism on this energy balance. We used a box model with six different zones that vary in latitude with boundaries at 60°S, 30°S, 0°, 30°N, and 60°N. While the nature of this box model is similar to that of the carbon cycle model we explored, this experiment aims to answer a different set of questions about an entirely new model.

We implemented the box model using a Runge-Kutta integration method. Since each zone spans different latitudinal regions, the ratios of land, water, and ice covering the Earth's surface vary from one zone to another. We quantified these fractional areas of land, water, and ice in each zone using Google Earth imagery. With these fractions we calculated the average albedo for each zone using the equation in Appendix B and values given in Tables 1 and 2 of the assignment. We also operated under the assumption that Earth wasn't a perfect black-body, hence why its emissivity is taken to be 0.95 throughout the paper.

With these values, we generated steady-state plots to which volcanic forcing was later added. The values for forcing were based on Figure 2 in Robock's paper (2000). We then ran experiments where albedo varies with temperature to observe a Snowball Earth scenario triggered by volcanism. Finally, we explored the relevance of the timing of eruption, i.e., its seasonality, by incorporating the tilt of the earth into all the individual zonal geometric factors.

RESULTS

The steady-state solution for the 6 zones is plotted in Figure 1. The x-axis is a 15 year time period and the y-axis is temperature in Celsius. To this, we added a volcanic forcing function, based on Figure 2 of the Robock [2000] paper. The function was created by interpolating the values for direct broadband radiation in Figure 2. The data for the DB radiation was derived from the Pinatubo eruption, so our figure presents the results for the same. The eruption was set at $t = 2$ years to ensure that the model was in steady-state prior to it. There is very minimal change seen in the temperatures of zone 1 and zone 2, with a very minor cooling seen in zone 2 4 years post-eruption. A noticeable lag of roughly 4 months seen in zone 6 post eruption while zones 4 and 5 show an immediate cooling response to the eruption. We also observe a return to the initial steady-state temperature values after ~ 10 years for all the zones, i.e., 8 years post eruption.

In Figure 1, albedo has constant values, i.e., it remains unchanged through time. Next, we parameterized albedo to be a function of temperature and plotted Figure 2. The x-axis is a 60 year time period with the eruption year set to $t=25$ and the y-axis is temperature in Celsius. Figure 2a uses exactly the same forcing values as Figure 1, with the albedo now being temperature dependent. The change in the nature of albedo is why there is a minute dip in the steady-state values for temperature across 2a, 2b and 2c. The temperature soon after obtains a new steady-state and stays constant there until a volcano is forced in at $t = 25$ years. The cooling effect is more pronounced in Figure 2 than in Figure 1. In 2a, much like Figure 1, the cooling response from zones 4 and 5 are instantaneous (no lag). Zones 2 and 1 show more cooling here than in Figure 1, with respective lag times of 8 months and a year. The model returns to steady-state after.

2b has the same values as 2a, save for the volcanic forcing. The forcing was doubled. A more pronounced cooling effect is seen in all zones but lag times appear to be unchanged. Like 2a, 2b returns to steady-state as well. 2c shows the forcing rate of 2b (double of Pinatubo), with double the residence time (4 years) as well. Upon eruption, temperatures drop across all zones and do not revert to the initial steady-state values and acquire new, significantly lower steady-state values. 2c shows a Snowball Earth simulation.

Figure 3 shows the temperature response of Zone 5 ($30-60^\circ$ N) for four different eruption scenarios where Pinatubo-like eruptions are simulated between year 1 (Dec. 21) and year 1.75 (Sept. 21). The temperature response is plotted for 8 years (6.25 - 7 years post eruption). In all cases the temperatures return to the pre-eruption steady-state. In each scenario the magnitude and location of the eruption is held constant while the timing is varied so that the eruptions correspond with the winter solstice, spring equinox, summer solstice, and fall equinox. Additionally a running mean is computed for each scenario representing average annual temperature. We find the largest reduction in temperature occurs following an autumn eruption and the smallest reduction in temperature occurs following a springtime eruption. The difference between absolute minimum temperature in the spring eruption scenario and fall eruption scenario is 0.42 Kelvin and the difference between minimum annually averaged temperature in these scenarios is 0.32 Kelvin.

Zone Number	Land, Water and Ice Fractions Considered
Zone 1 (90°S- 60°S)	L: 0.00 W: 0.52 I: 0.48
Zone 2 (60°S- 30°S)	L: 0.07 W: 0.93 I: 0.00
Zone 3 (30°S- 0°)	L: 0.15 W: 0.85 I: 0.00
Zone 4 (0°N-30°N)	L: 0.25 W: 0.75 I: 0.00
Zone 5 (30°N-60°N)	L: 0.50 W: 0.50 I: 0.00
Zone 6 (60°N- 90°N)	L: 0.35 W: 0.50 I: 0.15

TABLE 1: The land, water and ice fraction values for each of the zones as used in the code to produce the figures below. Based on data from Google Earth Imagery.

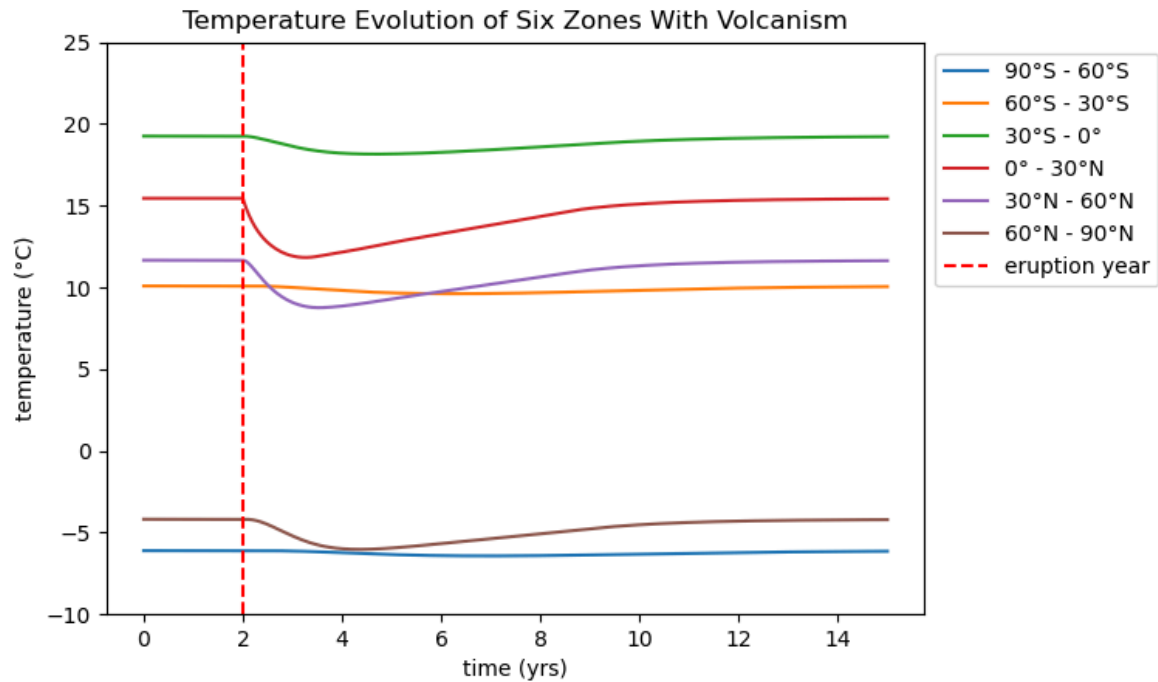


FIGURE 1: Steady-state solution of the six-zone model with volcanic forcing beginning at t=2 years, integrated over 15 years.

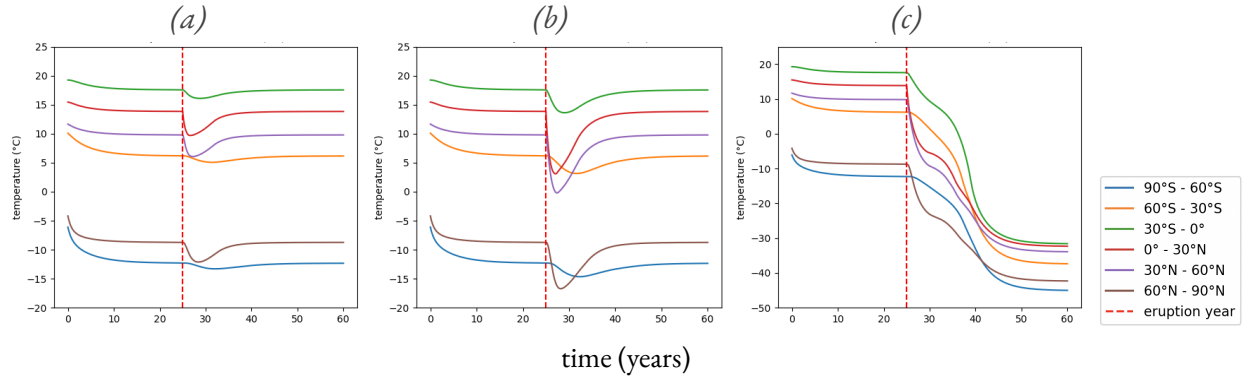


FIGURE 2: Steady-state solutions of the six-zone model with temperature-dependent albedo and varying volcanic events (2(a), 2(b), 2(c)), each integrated over 60 years. 2(a) represents the 1991 Pinatubo eruption, 2(b) represents an eruption with a forcing effect double that of Pinatubo's, while 2(c) represents an eruption with a forcing effect double that of Pinatubo's and whose forcing has double the residence time.

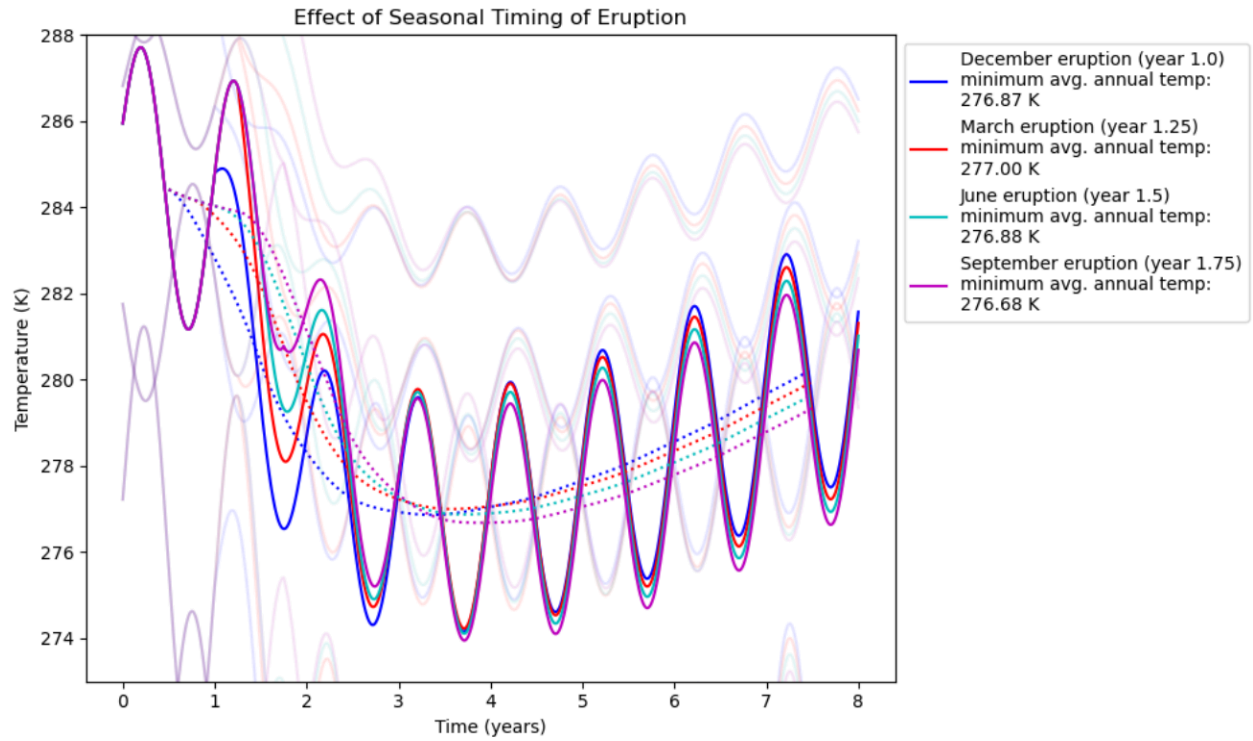


FIGURE 3: Impact of seasonal timing of Pinatubo-like eruptions (15° N) on temperature in Zone 5 ($30-60^{\circ}$ N) over the years following the event with annual mean temperature plotted as a dotted line.

DISCUSSION

Following an eruption, volcanic ash and dust in the stratosphere usher in a cooling effect by reducing the quantity of incoming solar radiation. Pinatubo is located at 15°N (that is located in zone 4), which is why zone 4 sees the quickest cooling, with no lag time, and zone 5 follows closely after with a lag time of a few days. The thermal exchange coefficient from zone 4 to 5 (K_{45}) is 5 times higher than that from zone 4 to 3 (K_{34}), explaining the reduced cooling observed in zone 3, south of the eruption. K_{45} is so significantly large owing to the presence of the North Atlantic jet stream, which steers currents and wind flows northwards, increasing advection of cold air from zone 4 to 5. The Pinatubo eruption had a residence time of 4 years (Robock [2000]). In Figure 1, up to 12 years post-eruption data is plotted. The choice to pick 3 times the residence time of Pinatubo was intentional so as to capture any minor discrepancies in temperature that may happen after the calculated residence time. This is also why zone 6 sees more of a lag, given its distance latitudinally from the eruption location.

A noteworthy point to be made about this figure is that the model runs colder than we expect the steady-state temperatures to be, this is likely due to the oversimplification of our model and its failure to encapsulate some effects such as zonal emissivity variation, greenhouse gas effects, etc.

Figure 2 shows how volcanism can induce a Snowball Earth scenario. We modeled this by varying albedo with temperature – between a threshold of 260° K and 290° K, our zonally-averaged albedos decreased quadratically with temperature until reaching a constant value for the albedo of ice, 0.6 (Equation 2, Appendix A). A 60-year time period was chosen to observe the steady-states. Only under the conditions in figure 2c did our model fall into Snowball Earth, whereas in 2a and 2b the models recover to their pre-eruption steady-states. Since the only difference between models 2b and 2c is a sustained aerosol residence time, we note that the magnitude of volcanic aerosols is not the only consideration to be made while modeling real-world eruptions; residence time is a pivotal component to consider as well.

In order to investigate the impact of the seasonal timing of a volcanic eruption we altered the geometric factor (gamma in Equation 1, Appendix A) to be dependent on time representing the 23.5° tilt of the earth. Geometric factor represents the ratio between solar radiation intercepted and the area where longwave radiation is emitted. This model combines time dependent geometric factors to represent axial tilt with temperature dependent albedo values that captures the seasonal variation of global surface temperatures. Like previous iterations this form of the model also runs cold. To prevent collapse into a ‘Snowball Earth’ scenario, temperature thresholds in the temperature dependent albedo function are reduced by 5 Kelvin.

Our finding that an Autumn eruption results in a 0.32 Kelvin reduction in minimum average annual temperature compared to a springtime eruption (Figure 3) shows that eruption timing does have an effect on the magnitude of cooling. This value is small but could still have widespread impacts such as reduced crop yields and increased deaths due to freezing temperatures. Additionally, in scenarios with larger eruptions the seasonal impact is increased and the variation in temperature related to eruption timing could mean the difference between crossing a threshold into an ice-albedo feedback loop leading to snowball earth or remaining in an interglacial steady-state.

CONCLUSION

In this study, we developed a simplified radiative energy balance model to examine the impact of volcanism on the Earth's climate system. Using a six-zone box model spanning different latitudes, we quantified land, water, and ice fractions to calculate zonal albedo and incorporated real-world parameters such as emissivity and volcanic forcing. Our findings illustrate the complex interactions between volcanic eruptions and global temperatures, highlighting the significant cooling effect caused by volcanic aerosols in the stratosphere. This cooling effect was most pronounced in regions close to the eruption site, with temperature discrepancies and lag times explained by thermal exchange coefficients and atmospheric circulation patterns.

Our model demonstrates that volcanism has the potential to trigger extreme climate events, such as a Snowball Earth scenario, particularly when aerosol residence times are extended. By simulating temperature-dependent albedo and including the Earth's axial tilt, we captured the influence of seasonal variability on post-eruption climate responses. The effect of eruption timing was evident, with autumn eruptions causing a greater reduction in minimum average annual temperatures compared to spring eruptions.

Overall, our results emphasize the importance of understanding both the magnitude and timing of volcanic eruptions in climate models. The interplay between volcanic aerosols, surface albedo, and seasonal variations plays a critical role in determining the extent of cooling and the risk of entering a feedback loop that would result in a Snowball Earth state. Future research should focus on expanding the model to more accurately represent earth's climate system with terms to represent the spread of aerosols, impacts on ozone, and the presence of long lasting greenhouse gasses.

CONTRIBUTIONS

Josh: Wrote the results and discussion for figure 3 and the code for it as well as the conclusion.

Ethan: Wrote the body/bulk code for the assignment, code for figure 1 and 2 and the discussion for figure 2 as well.

Shreya: Wrote the introduction, conclusion, table 1, results for figures 1 and 2 and discussion for figure 1. Also assisted with the code.

APPENDIX A

$$\frac{dT_i}{dt} = \frac{1}{\rho_i c_i [Z_i]} \left\{ \gamma_i \left(1 - \alpha_i^{\text{sky}} \right) (1 - \overline{\alpha_i}) S_0 - \tau \sigma_B T_i^4 \right\} + \frac{1}{A_{i-1} \rho_i c_i [Z_i]} \left\{ -L_{i,i-1} k_{i,i-1} (T_i - T_{i-1}) + L_{i,i+1} k_{i,i+1} (T_{i+1} - T_i) \right\}$$

Equation (1)

$$\alpha(T) = \begin{cases} \alpha_i & T \leq T_i \\ \alpha_o + (\alpha_i - \alpha_o) \frac{(T - T_o)^2}{(T_i - T_o)^2} & T_i < T < T_o \\ \alpha_o & T \geq T_o \end{cases}$$

Equation (2)